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PROJECT TITLE:

DUAL CORE EDGE COMPUTING FRAMEWORK FOR POWER FACTOR CORRECTION AND LOAD DEMAND MANAGEMENT

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ABSTRACT

Managing fluctuating load demand and maintaining near-unity power factor presents significant challenges. This Dual Core Power Factor Optimization and Smart Load Demand Management System integrates with real-time power quality monitoring to achieve ≥0.98 power factor and 15-25% peak demand reduction through predictive load shifting and intelligent demand response. Featuring reinforcement learning-based demand response, neural network compensation sizing, precise electrical parameter sensing, and adaptive reactive power control via capacitor banks or STATCOM, the IoT-enabled edge computing solution delivers harmonic mitigation and substantial cost savings. Scalable for industrial and utility applications, it overcomes traditional APFC limitations through predictive analytics and smart demand orchestration for sustainable power systems.

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INTRODUCTION

- Rapid technological advancement and growing electrical energy demand require innovative power management solutions
- Traditional frameworks struggle with modern challenges like variable consumption trends and renewable energy integration
- Need for real-time, adaptive control systems in contemporary power grids
- Smart Integrated Load and Power Management System ensures a groundbreaking approach transforming electrical power management
- Combines established load compensation methods with IoT and technologies
- Self-regulating control system providing high-precision, real-time Power Factor optimization
- Function Fitting Neural Network architecture chosen over traditional methods
- Moves beyond slow rule-based systems and evolutionary algorithms (e.g., Genetic Algorithms)

INTRODUCTION (Continued..)

- Operates under Supervised Learning framework for intelligent decision-making
- Automatically maps hidden patterns between varying loads – non-linear loads
- Perfect Power Factor Control: Maintains 1.0 PF via Capacitor Bank
- Handles complex grid data in milliseconds on ESP32 microcontroller
- Optimizes renewable sources (Solar and Wind) based on real-time load spikes and predictive weather patterns
- Improves efficiency, reliability, and sustainability of contemporary power systems
- Cohesive, self-regulating control with groundbreaking performance

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LITERATURE REVIEW

Author	Year	Adopted scheme	Features	Challenges
Biswas et al.	2025	AI frameworks for smart grid optimization including demand response, renewable forecasting, and real-time control	Integration strategies for, IoT, and power quality management across grid scale operations	Broad review lacks specific implementation details for industrial PF correction and capacitor bank control hardware integration
Witharama et al.	2024	Advanced Genetic Algorithm with LightGBM forecasting for multi-objective microgrid optimization	Comprehensive cost modeling including battery degradation and demand response coordination for long-term optimization	Computationally intensive GA unsuitable for real-time PF correction requiring sub-second capacitor switching decisions
Jat et al.	2024	AI/ML controllers (neural networks, fuzzy logic) for PV grid stabilization	Real-time fluctuation mitigation and reactive power management for renewable integration	Solar PV-specific focus requires adaptation for general industrial load PF correction scenarios
Farsi et al.	2021	Parallel Deep LSTM-CNN hybrid model for short-term load prediction	Superior temporal pattern capture and spatial feature extraction for accurate demand forecasting	High computational complexity of parallel architecture

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LITERATURE REVIEW

Author	Year	Adopted scheme	Features	Challenges
Mishra et al.	2020	Dynamic load balancing algorithms for resource optimization	Multi-criteria load distribution strategies adaptable to power system demand management scenarios	Cloud computing with real-time electrical power constraints
Swarna Priya et al.	2020	Hybrid Wind Driven Optimization (WDO) + Firefly Algorithm for IoE energy cloud clustering and CH selection	Bio-inspired optimization for energy-efficient load distribution across distributed IoT networks	Energy cloud IoE focus incompatible with single-site industrial PF correction requiring deterministic real-time capacitor switching

Proposed Solution Summary

Our project overcomes these challenges by replacing slow cloud reliance and complex algorithms with deterministic edge-computing. Using a dual-core ESP32, our system executes lightweight Neural Networks locally, guaranteeing the sub-second capacitor switching and precise hardware integration required for reliable, single-site industrial Power Factor correction and load management.

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EXISTING SYSTEM

- **Fixed Capacitor Switching -** Manual or relay-based control unable to adapt to load variations
- **Overshoot & Resonance -** Inadequate compensation causes system instability and inefficiency
- **Utility Penalties -** Fixed tariffs impose fines for low power factor regardless of real-time conditions
- **Reactive Power Losses** - Additional transmission losses and increased kVA demand charges
- **Uncoordinated Control -** Power factor correction and demand response operate independently without integration
- **No Real-Time Monitoring -** Lack of visibility prevents early fault detection and proactive adjustments

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PROPOSED SYSTEM

- **Neural Network Analysis** – Real-time power factor monitoring and adaptive capacitor bank control
- **Near-Unity Power Factor** – Dynamic maintenance across varying loads, eliminating overshoot & resonance
- **IoT Sensing Integration** – Continuous electrical parameter monitoring with edge computing
- **Reinforcement Learning** – Smart demand response automatically shifts flexible loads
- **Real-Time Optimization** – Responds to tariffs, grid signals, and operational constraints
- **Predictive Analytics** – Harmonic mitigation and fault prevention through forecasting
- **Continuous Feedback Loops** – Self-optimizing performance adapts to load changes
- **Renewable Integration Ready** – Handles variable generation and distributed energy resources
- **Proven Results** – Eliminates penalties, 15-25% peak demand reduction, sustained power quality
- **Interface for users and admin** – Web application accessed by the users and the admins

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HARDWARE/SOFTWARE REQUIREMENTS

PREREQUISITES	SOFTWARE	HARDWARE
Knowledge of Power Systems & Power Electronics	MATLAB / Simulink	Microcontroller: ESP32, Transformers
Understanding of Control Systems & Automation	Cloud Platform (AWS)	Relays & Motor Drives, Power Supply Units, Capacitor Banks
Basics of Machine Learning & Neural Networks	Supervised Learning	Data Logger & Network Switch
IoT & Embedded Systems Concepts	LabVIEW	IoT Module: Wi-Fi, MQTT

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COMPARISON OF CONTROLLERS

Feature	Arduino Uno	Raspberry Pi	ESP32 (Chosen)	Justification
Processor Speed	16 MHz	1.5 GHz	240 MHz (Dual-Core)	Handles the 0.0002s logic cycles that Arduino cannot process.
Core Architecture	Single-Core	Quad-Core	Dual-Core (Symmetric)	Core 0 runs PF Optimization while Core 1 manages Load Demand.
Built-in Connectivity	None (Requires Shields)	Wi-Fi / Bluetooth	Integrated Wi-Fi / BLE	Direct uplink to the User/Admin Web App without extra hardware.
ADC Resolution	10-bit	N/A (Digital only)	12-bit (High Precision)	Provides the precise current/voltage data needed for the Neural Network.
Power Consumption	Medium	Very High	Low Power (Sleep Modes)	Fits the project goal of an energy-efficient, self-regulating control system.
Cost & Complexity	Low / Simple	High / Complex OS	Optimized Efficiency	Offers the "Edge AI" power of a Pi with the hardware simplicity of an Arduino.

METHODOLOGY

The approach of the Smart Integrated Load Demand Management System incorporates a control circuit, along with load monitoring units associated with both the load and generation aspects. Power generated by motors paired with alternators is regulated through IoT-enabled Narrow Band communication, providing voltage-adjusted power to users according to immediate demand. An overload is detected by a current transformer, which triggers the controller to engage additional motors for supplementary power, repeating this process as necessary.

Prior to providing extra power, the synchronization unit guarantees the alignment of voltage and frequency. The system independently handles load requirements, ensuring a consistent and dependable power supply, while also accommodating various renewable energy sources. The project progression consists of acquiring components, conducting tests, designing, developing, assembling, and monitoring.

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METHODOLOGY – TECHNICAL SPECIFICATIONS

- Dual-Core Processing (ESP32) Core 0 executes Neural Network PF Correction (200µs rate) and Core 1 manages Source Switching and MQTT/IoT data.
- Multi-Layer Perception (MLP) Neural Network maps P, Q, and V to maintain 1.0 PF via a Static VAR Compensator.
- Load Demand Management ensures Predictive Source switching for Solar (>6kW) and Wind (>9kW) with a 500W hysteresis buffer.
- ACS712 sensors with 12-bit ADC provide 0.8mV/unit resolution for real-time current monitoring.
- Discrete solver at 0.0002s sample time successfully tests 4kW to 15kW load-step transitions.
- Sensor data streams via MQTT to a Firebase Database for Admin/User web-interface monitoring.

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REAL-TIME INTEGRATION

- A dual-core ESP32 Microcontroller serves as the central "Brain," executing two independent algorithms in parallel to ensure zero-latency grid management. The system uses a Multitasking RTOS (Real-Time Operating System) environment to separate the Power Factor (PF) optimization from the Load Demand logic.

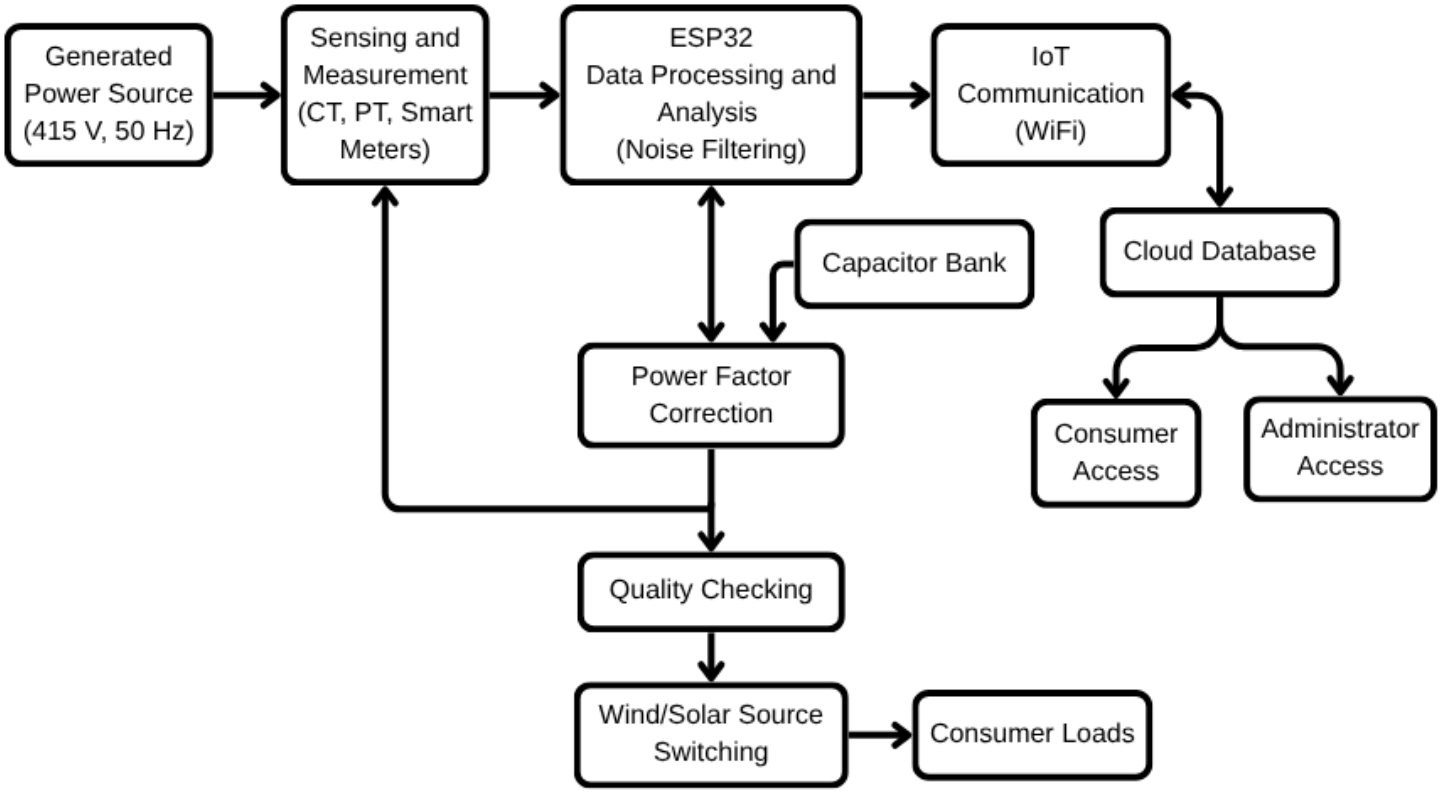
- Function Fitting Neural Network (Supervised Learning) is used for the Real-time processing of Grid P and Q data to calculate the exact reactive power injection required for a 1.0 Power Factor. It Controls the Static VAR Compensator through high-speed Pulse Width Modulation (PWM).

- Predictive Logic & Threshold Analysis is used to Monitor the P-load and Q-load of the Heavy Load branches while cross-referencing real-time Weather API data. It automatically triggers Solar and Wind source breakers to prevent grid overloading during peak consumption events.

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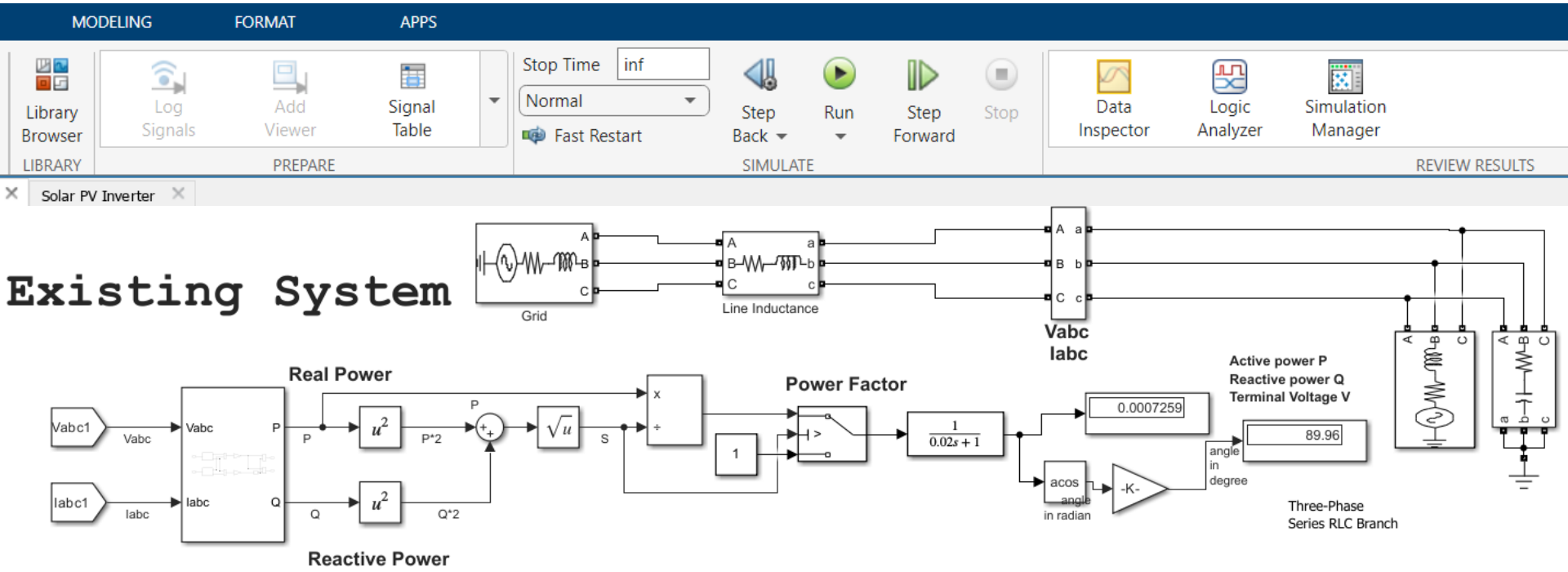
WORKFLOW



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EXISTING SYSTEM SIMULATION



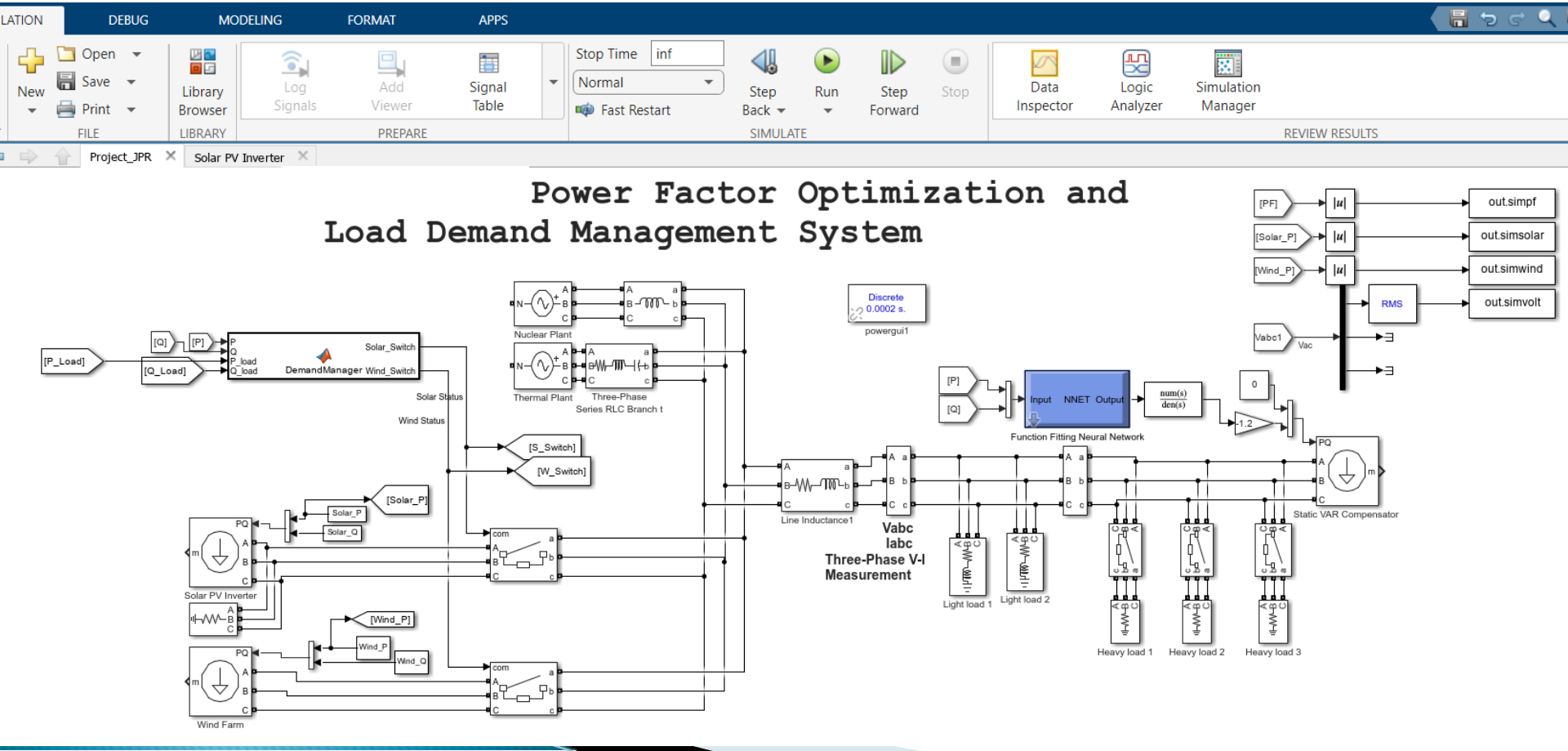
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POWER FACTOR OPTIMIZATION AND LOAD DEMAND MANAGEMENT SYSTEM



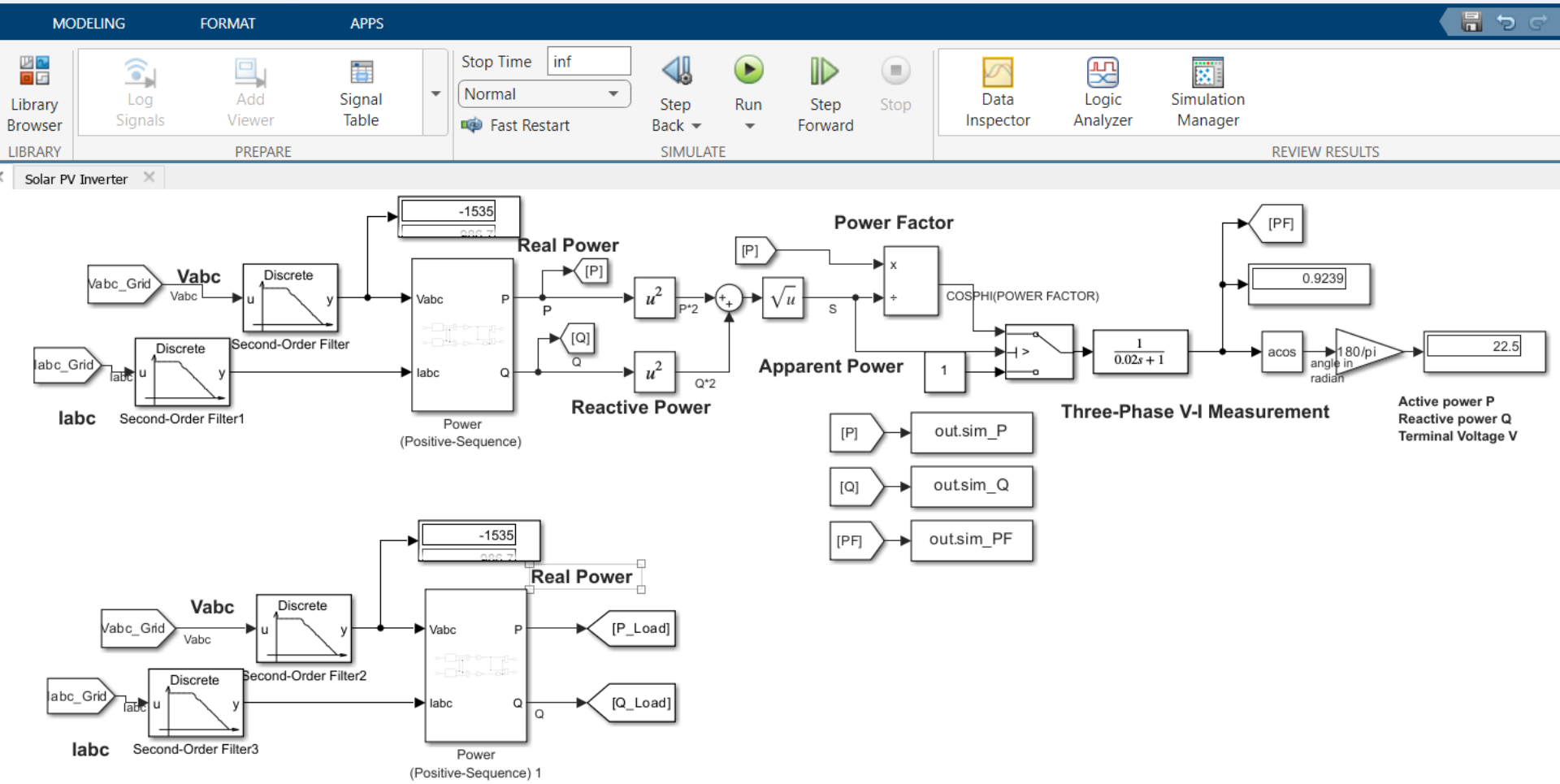
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POWER FACTOR ANALYSER



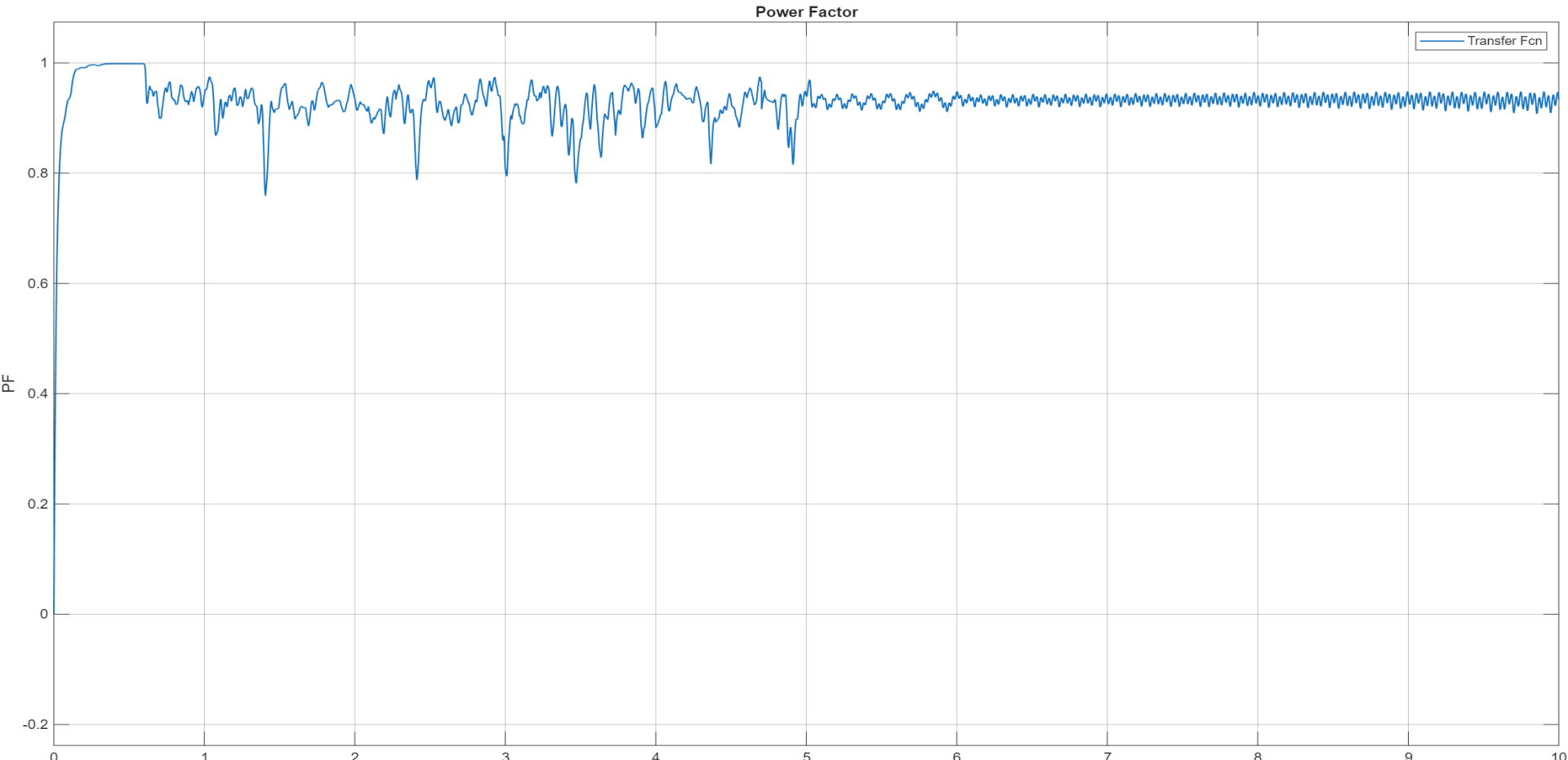
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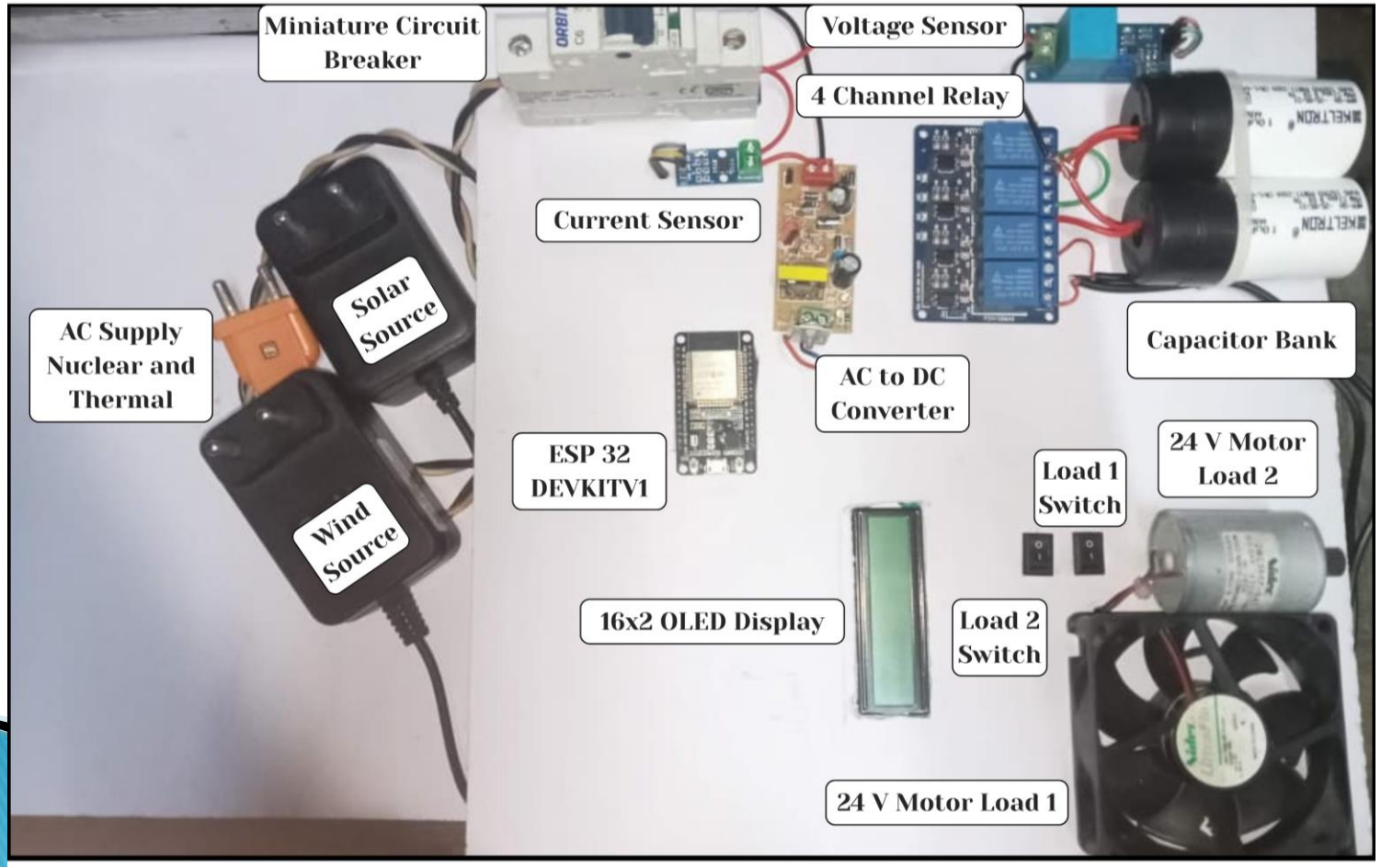
RESULT



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HARDWARE PROTOTYPE



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CONCLUSION

Thus, our project successfully demonstrates a groundbreaking approach to contemporary power management by transitioning from traditional cloud-reliant systems to a deterministic, **dual-core edge computing framework**. By utilizing the parallel processing capabilities of the **ESP32**, the system effectively decouples high-speed **neural network-based Power Factor optimization** (Core 0) from **intelligent load demand management** and IoT telemetry (Core 1). The hardware prototype achieved its primary engineering goals like maintaining a **near-unity power factor** to eliminate utility penalties and ensuring an uninterruptible, **"always-on" microgrid** through a diode OR-ing bus that seamlessly integrates renewable solar and wind sources during AC mains failure. Ultimately, this self-regulating IoT solution delivers a scalable model for industrial efficiency, providing verified results of **15-25% peak demand reduction** and sustained power quality for sustainable electrical systems.

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THANK YOU